

Intelligent agents

An introduction to artificial intelligence

Intelligent Systems

School of Computer Science · Ciudad Real

Rubén Juárez Cádiz



A route towards agent design

Learning outcomes and classroom use

01 · Foundations What AI means and how an agent interacts with its environment.

02 · Decisions Rationality, utility, performance measures and constraints.

03 · Design PEAS, environment properties and agent architectures.

04 · Applications LLM-based agents and the geared Loopover laboratory.

By the end, you should justify what an agent must perceive, remember, decide and learn.

The robot arrives. Was its action good?

Running example · An autonomous campus delivery

It delivers the parcel in 4 minutes.

It crossed a prohibited area.

What else do we need to evaluate its decision?

Four approaches to artificial intelligence

Thinking / acting · Humanly / rationally

	Human perspective	Rational perspective
Thinking	Model human cognitive processes.	Derive conclusions using formal rules.
Acting	Produce behaviour comparable to human behaviour.	Choose actions that support a goal using the available information.

This course mainly adopts the rational-agent perspective.

The Turing Test and its limits

Observable behaviour

The imitation game compares responses through written conversation.

Conversational test

Language processing, knowledge representation, reasoning and learning.

Total version

Adds perceptual and physical interaction, such as vision and robotics.

Convincing conversation does not certify safety, truthfulness or competence at every task.

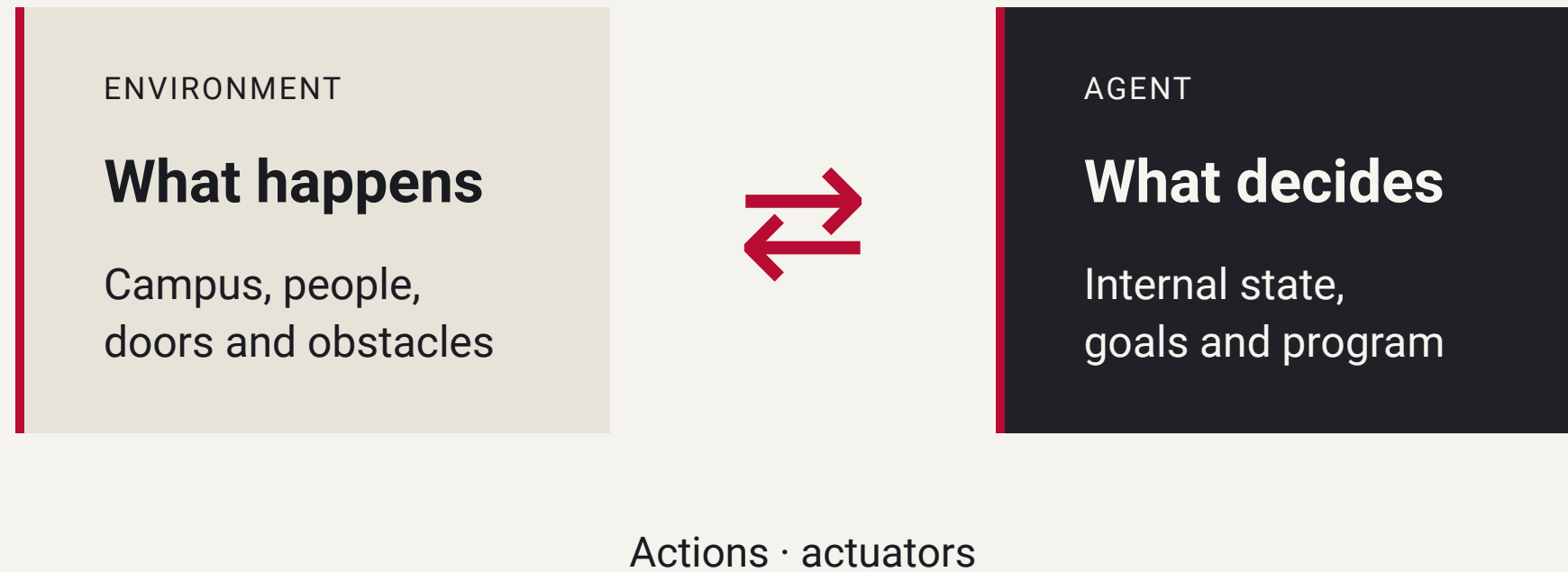
Ideas accumulate

1950	1956	1980s	2012	2023
Imitation	Dartmouth	Expert systems	Deep learning	ReAct
Observable behaviour and conversation.	The field acquires its name.	Explicit knowledge and rules.	Major progress in visual recognition.	Language models interleave reasoning and actions.

Intelligent agents predate LLMs. Search, logic, learning and control continue to work together.

The basic relationship: agent and environment

Observations · sensors



An observation informs the agent. An action may change the world and future observations.

Function, program and architecture

$$f : P^* \rightarrow A$$

P*: all finite percept histories.

A: the set of possible actions.

The function specifies an action for each history.

Function

An abstract description of behaviour.

Program

An implementation that computes an action.

Architecture

The platform that runs the program and connects inputs and outputs.

Agent = architecture + program

State, observation and belief

World state · s_t

The door is closed and someone is behind it.

Observation · o_t

The camera only sees the door.

Belief · b_t

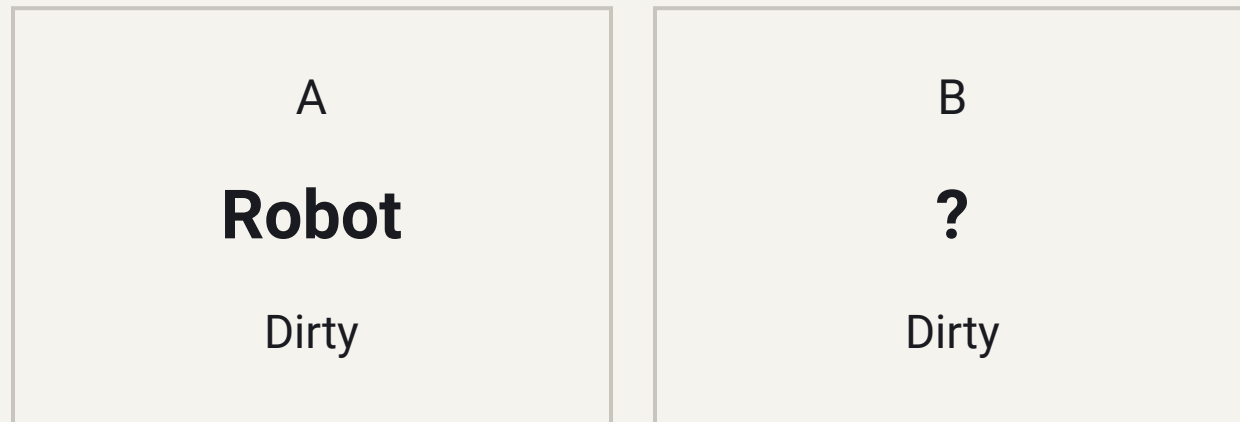
Several hidden states remain possible.

$$b_t(s) = P(s_t = s \mid h_t)$$

A probabilistic belief summarizes which states are plausible given the history of observations and actions.

Partial observation may require memory or inference.

A small world with real decisions



Percept

Current location and local dirt status.

Actions

Left, right, suck and wait.

Assumptions

Two squares. Accurate sensors. Dirt does not return.

Challenge

Clean the world while avoiding unnecessary actions.

The lecturer sees both squares. The agent only senses its current square.

A perception and memory laboratory

Square A

Robot

Dirty

Square B

No robot

Dirty

Execute one step

Reset

Agent program

Simple reflex ▾

World view for the observer. The agent receives only its local percept.

Available percept

A · dirty

Agent memory

No memory retained

Last action / steps

Not executed · 0 steps

When can the agent justify choosing “wait”?

Memory changes later actions

Step	Simple reflex	With memory
1	Suck in A	Suck in A
2	Move to B	Move to B
3	Suck in B	Suck in B
4	Move to A	Wait
5 and 6	Keep alternating	Wait

If movement consumes energy, the same cleanliness with less movement can improve performance.

Rationality: deciding with what is available

Choose the action with the best expected performance, given the evidence and the agent’s knowledge.

01	02	03	04
Criterion	Knowledge	Options	History
What matters	What is known	What can be done	What has been observed

A good decision can still fail

TWO ROUTES · HYPOTHETICAL EXAMPLE

Route	Outcome	Utility
Safe	Always arrives	+6
Fast	80% success	+10
Fast	20% failure	-8

EU(fast)
= 0.8 · 10 + 0.2 · (-8)
= 6.4

EU(safe) = 6

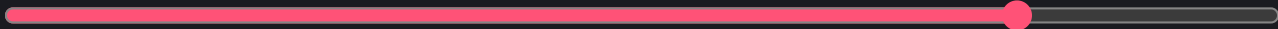
The fast route has higher expected utility, although it may fail.

Utilities encode the preferences in this example. They are neither minutes nor percentages.

When does the decision change?

Success probability of the fast route

80%



Success: +10 · Failure: −8

Safe route: +6

Adjust the probability and observe the recommendation.

The indifference threshold is $p = 14/18 \approx 77.8\%$.

Fast

6.4

Safe

6.0

Higher expected utility: fast route.

Bar length: EU + 8 on a common scale from 0 to 18. The numbers show expected utility.

The metric shapes behaviour

Isolated metric	Possible unintended behaviour	More complete evaluation
Alerts closed	Close alerts without investigation.	Confirmed detection, errors and response time.
Tutor messages	Prolong the conversation.	Learning on a new task and quality of support.
Parcels per hour	Take inappropriate shortcuts.	Verified deliveries within safety constraints.

External performance, training reward and internal utility can differ.

Preferences and constraints

Preferences with trade-offs

Shorter delivery time.

Lower energy consumption.

Greater comfort.

maximize $E[U]$

Mandatory conditions

Respect authorized access.

Avoid excluded areas.

Stop if a risk remains unresolved.

$a \in A_{\text{allowed}}$

A high reward does not make a prohibited action permissible.

PEAS: specifying the task

P

Performance

How is success evaluated?

E

Environment

What world does it operate in?

A

Actuators

What can it change?

S

Sensors

What information does it receive?

PEAS describes the task. An architecture explains how the agent tackles it.

PEAS for the delivery robot

P · Performance Correct delivery, time and energy. Prohibited access is a constraint.

E · Environment Buildings, routes, pedestrians, doors and delivery points.

A · Actuators Motors, brake, indicators and delivery notification.

S · Sensors Camera, proximity, localization and battery status.

Scenario assumption: the robot receives an authorized destination and may stop or request help.

Activity: a tutor for this course

6 min

In pairs, specify PEAS and suggest a metric that could encourage unwanted behaviour.

Deliverable: four rows and a justification.

One possible answer

P

Learning on a new problem and correctness.

E

Student, materials and conversation.

A

Explain, give hints and request a check.

S

Student responses, history and documents.

Counting quick answers may reward solving the task without helping the student learn.

The environment shapes the design

Before choosing an algorithm,
we need to describe the problem.

Observability · Uncertainty · Temporal dependence

Dynamics · Representation · Other agents · Knowledge

Observing and knowing are different questions

Dimension	Question	Design implication
Observability	Can I observe all relevant state information?	Missing information may call for memory or inference.
Determinism	Do the state and action determine the next state?	Multiple possible outcomes require uncertainty modelling.
Known / unknown	Do I know the transition rules or probabilities?	Unknown rules must be estimated or learned.

The next roll of a fair die has an uncertain outcome and a known distribution.

Time, representation and other agents

Dimension	Contrast	Example with explicit assumptions
Temporal dependence	Episodic / sequential	Independent image labels / route planning.
Change during deliberation	Static / dynamic / semidynamic	Untimed puzzle / traffic / timed chess.
Representation	Discrete / continuous	Grid squares / real-valued position and speed.
Interaction	Single-agent / multi-agent	Solo puzzle / a game with an opponent.

Discrete spaces may be finite or countably infinite. A task may mix discrete and continuous variables.

Classification depends on assumptions

Property	Classroom Loopover	Timed chess	Campus robot
Observation	Full	Full ¹	Partial
Transition	Deterministic	Deterministic ²	Stochastic in the model
Decisions	Sequential	Sequential	Sequential
During deliberation	Static, no clock	Semidynamic	Dynamic
Representation	Discrete	Discrete + clock	Mixed
Agents / rules	One / known	Several / known	Several / partly known

¹ Game state and clock are available. ² The rules determine each move’s effect. The opponent introduces strategic uncertainty.

One changed assumption changes the agent

Visible puzzle

Hidden tiles

Noisy action

Unknown rules

The entire board is visible and action effects are known.

Observation	Full
Transition	Deterministic and known
Design requirement	Search over the available state.

A classification should explain what the agent needs in order to decide.

Four decision mechanisms

Architecture	Main information	Question answered
Simple reflex	Current percept and rules	Which action matches the situation now?
Model-based	Memory and a world model	Which state explains what I observe?
Goal-based	Model and goal condition	Which action sequence can reach the goal?
Utility-based	Preferences and consequences	Which option has the greatest expected value?

Learning can improve any of these architectures. It is not an obligatory final rung.

Simple reflex agents

01

Perceive

Current input

02

→ Match

A rule condition

03

→ Act

Associated action

```
if dirt == DIRTY:
    return SUCK
if location == A:
    return RIGHT
return LEFT
```

Strength

Little computation and a fast response.

Limitation

Situations that look the same receive the same response even when their histories differ.

Model-based agents

$$m_t = \text{update}(m_{t-1}, a_{t-1}, o_t)$$

Previous memory

What had been observed or inferred.

Previous action

Which change was attempted.

New observation

Which evidence has now arrived.

In the vacuum world

The agent remembers that A is clean while it is in B.

If the world changes

Memory may become stale and require revision.

A model supports state estimation. It does not guarantee that the estimate is correct.

Goal-based agents

Goal

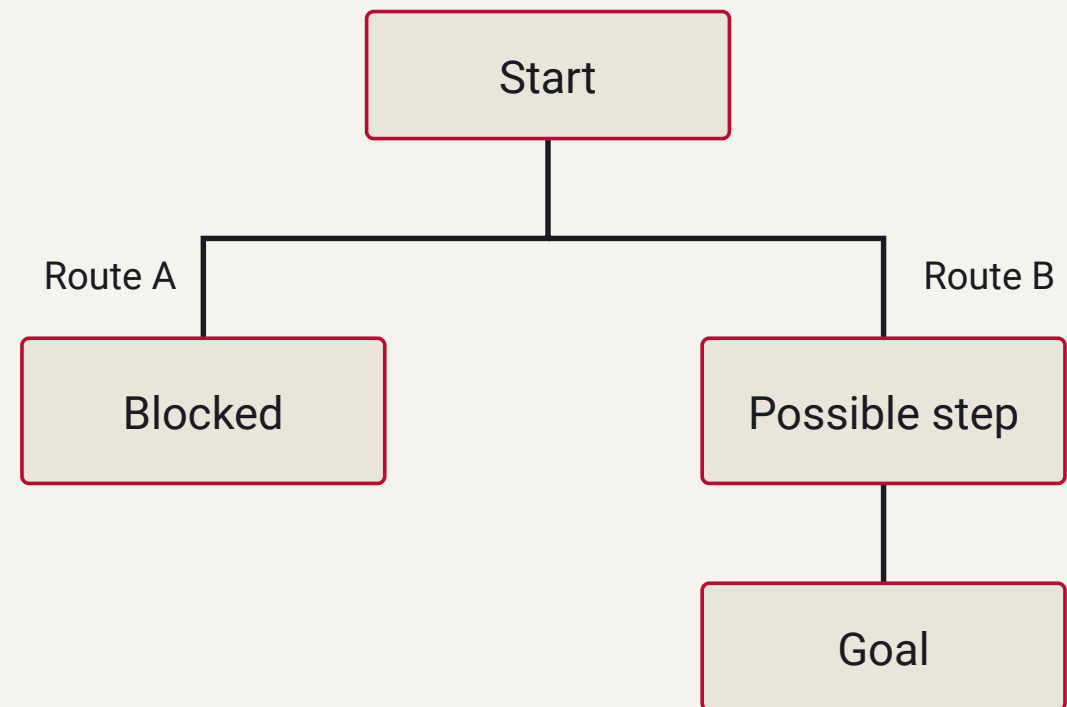
The parcel is at the authorized destination.

Plan

A sequence of actions that reaches that condition.

Replanning

A new observation can invalidate the route.



Search simulates consequences before acting.

Utility-based agents

Feasible plan	Normalized time t	Normalized energy e	$U = -(0.7t + 0.3e)$
A	0.2	0.8	-0.38
B	0.5	0.3	-0.44

With these weights

A has greater utility and is preferred.

When energy matters more

With $U = -(0.3t + 0.7e)$, B wins:
-0.36 versus -0.62.

Weights express preferences. Normalization avoids adding incompatible scales.

Learning improves behaviour

PERFORMANCE ELEMENT

Selects actions

Uses the current behaviour.

Critic	Evaluates experience against the criterion.
Learning element	Updates behaviour or its supporting knowledge.
Problem generator	Proposes informative experiences.

Exploration can provide useful information, but it also has a cost.

Decision-making also uses resources

A solution that arrives too late may no longer be useful.

Bounded rationality

Time, memory and computational capacity constrain deliberation.

Value of information

More observation is worthwhile when its expected benefit exceeds its cost.

Example: checking a door before planning the entire route may prevent a detour.

The language model within an agent

01

Observation

Request and state

02

→ Decision

Select a next step

03

→ Tool

Execute an action

04

→ Result

Verify and update

The LLM may propose

A query, a response or a structured action.

The system must control

Permissions, arguments, execution, limits and result verification.

ReAct studies the combination of expressed reasoning, actions and new observations.

Fixed workflow and dynamic selection

Aspect	Predefined workflow	Agent with dynamic selection
Step order	The design fixes transitions.	Depends on runtime observations and decisions.
Strength	Predictable execution of a bounded process.	Adaptation when required steps change.
Challenge	Exceptions must be modelled.	It may select unnecessary or incorrect steps.
Hybrid design	Mandatory controls and known operations.	Limited choice among authorized actions.

Additional autonomy needs a concrete purpose and an evaluation of its outcome.

Tool results can also fail

Untrusted input	An external document may contain instructions unrelated to the task.
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Invalid action	Malformed arguments or insufficient permissions may block execution.
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Uncertain result	A timeout does not establish whether the action occurred.
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Premature completion	A convincing answer may hide an unverified objective.
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	For the tutor: consult a source, check the evidence and acknowledge what cannot be verified.
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Evaluate trajectories as well as answers

What to assess	A classroom evaluation approach
Task success	Verify the final condition rather than accept a success claim.
Constraints	Count unauthorized actions or prohibited states.
Cost	Measure time, calls and resources under a shared budget.
Robustness	Repeat with varied inputs and controlled failures.
Generalization	Use held-out cases that were not used to tune the system.

Fair comparisons share tasks, conditions, budgets and evaluation criteria.

The actual laboratory: geared Loopover

00	01	02	03
04	05	06	07
08	09	10	11
12	13	14	15

A toroidal 4×4 board

No empty square. Tiles leaving one edge reappear at the opposite edge.

A coupled action

02+ shifts row 0 right, then column 2 down.

The second transformation uses the column index specified in the action.

One action, two transformations

00	01	02	03
04	05	06	07
08	09	10	11
12	13	14	15

Completed actions: 0

Row

0

Column

2

Sign

+

1 · Move row

Reset

Mixed example

Ordered goal. Action 02+ starts with row 0.

Linear state: 32 characters

00010203040506070809101112131415

The demonstration splits a legal action into two phases. The intermediate board is not a complete successor.

The 02+ transition, step by step

Initial state

00	01	02	03
04	05	06	07
08	09	10	11
12	13	14	15

1 · Row 0 right

03	00	01	02
04	05	06	07
08	09	10	11
12	13	14	15

2 · Column 2 down

03	00	14	02
04	05	01	07
08	09	06	11
12	13	10	15

Successor state: 03001402040501070809061112131015

Formulating the search problem

Component	Example specification
State s	A permutation of 00...15, encoded row by row in 32 characters.
Initial state s_0	The valid configuration received by the program.
Actions $A(s)$	Row-column pairs and a sign: $4 \times 4 \times 2 = 32$ actions.
Transition $T(s,a)$	Rotate the row, then the specified column.
Goal $G(s)$	$s = 00010203040506070809101112131415$.
Cost $c(s,a)$	Here, one unit per complete coupled action.

16! $\approx 2.09 \times 10^{13}$ bounds the permutation space. It is not a count of reachable states.

Search does not execute every path

State-space graph

Different action sequences may reach the same state.

Search tree

Represents explored paths and may repeat states.

Consequence

Recording visited states can prevent redundant work.

In the laboratory

Correct transitions come before evaluating a search algorithm.

For coupled actions, changing + to – without reversing the order need not undo the action.

Workshop: a cybersecurity agent

An agent reviews access alerts for a university laboratory. It can query logs, request context and propose a response.

10 min

Teams of three.
A one-page design.

1 • Specify

Complete PEAS and one constraint.

2 • Characterize

Justify observability, dynamics and other agents.

3 • Design

Choose an architecture and necessary memory.

4 • Evaluate

Define a normal case and a failure case.

Assumption: blocking an account requires authorization. Logs may be incomplete.

A reasoned workshop solution

PEAS	P: confirmed incidents and false positives. E: accounts and logs. A: query and propose. S: alerts and context.
Environment	Partially observable, sequential and dynamic. Other actors may change their behaviour.
Design	Per-alert memory, associated evidence and review status. Rules or utility depending on scope.
Evaluation	Confirmed alert, false positive, missing record and unauthorized request.

Suggested rubric: PEAS 3 points · environment 2 · architecture and memory 3 · evaluation 2.

Six questions to check understanding

01

What does P^* represent?

All finite percept histories, not just the current percept.

02

Can a rational decision fail?

Yes. It is assessed by expected performance given the available information.

03

Does unknown mean partially observable?

No. Unknown rules and hidden state are different difficulties.

04

Must a goal-based agent learn?

Not necessarily. It may plan with a supplied model.

05

What does utility add to a goal?

It compares the quality of different acceptable outcomes.

06

How many phases does 02+ have?

Two: row 0 right, then column 2 down. Together they count as one action.

Misconceptions we can now correct

Incomplete or incorrect claim	Correction
“Success proves rationality.”	Evaluate the choice using information available before the outcome.
“Every agent needs an LLM.”	Rules, search and other methods also implement agents.
“Learning means remembering.”	Storing information does not necessarily improve a rule or model.
“All discrete spaces are finite.”	A discrete set may be countably infinite.
“A goal is enough to compare plans.”	A goal tests achievement. Cost or utility compares alternatives.

Essential vocabulary

Term	Meaning
Percept	An input received by the agent at a particular instant.
Agent function	A mapping from percept histories to actions.
Policy	A rule for selecting actions from represented information.
Belief state	A representation of possibilities about hidden state.
Utility	A value representing preferences over outcomes.
Transition model	A description of the effects of actions on state.
Goal test	A predicate checking whether the objective is satisfied.

Readings for consolidation and extension

Course foundation

01_Presentacion_Tema1_Intelligent_Agents(2).pptx, including its visual specification of geared Loopover.

Russell and Norvig · Artificial Intelligence: A Modern Approach

Chapters 1 and 2: agents, rationality, PEAS and architectures.

aima.cs.berkeley.edu

UC Berkeley · CS188, Agents

A concise overview of agents and environments.

[CS188 · Agents](#)

Sutton and Barto (2018) · Reinforcement Learning: An Introduction

Second edition. Further reading towards the reinforcement learning unit.

[The authors' book page](#)

A modern reading and a closing question

Yao et al. (2023)

ReAct: Synergizing Reasoning and Acting in Language Models.

ICLR 2023. Initial preprint: 2022.

[Original paper](#)

[Authors' project and examples](#)

Reading question

Which part belongs to the model, and which depends on interaction with the environment?

Next unit

Formulating problems and searching for action sequences that reach a goal.

We can now assess the opening robot: goal, information, actions, constraints and consequences.